TIMOTHY NYACHIO

PROFILE

Software and Platform Engineer with 10+ years of experience across backend and frontend development, distributed systems, cloud infrastructure, DevOps/SRE, and technical leadership. Proven experience designing and operating scalable production platforms, leading cloud-native modernization and architecture improvements, standardizing observability and deployment practices, and driving infrastructure cost optimization. Strong background in Python, Go, C#, JavaScript/TypeScript, AWS, Kubernetes, Terraform, CI/CD, APIs, and distributed systems. Experienced in mentoring engineers, collaborating across Engineering, Platform, DevOps/SRE, and business teams, and improving reliability, developer experience, delivery standards, and operational efficiency across complex production environments.

PROFESSIONAL EXPERIENCE

Cloud-Native Software Engineer

ENGIE Energy Access (Africa) **Jan 2025 – May 2026**

- Designed, provisioned, and operated a production-grade **AWS EKS platform** hosting microservices and a centralized API gateway, improving availability, secure traffic routing, and workload isolation across multi-tenant environments.
- Implemented and extended **Apache APISIX on Kubernetes**, developing custom Lua plugins for user authorization and service-to-service authentication using digital signatures, strengthening API security and standardizing authentication across services.
- Deployed and configured **Istio service mesh** to improve service-to-service reliability through mTLS, traffic management, fault handling, and secure workload communication.
- Built and maintained **GitOps-based CI/CD workflows using Argo CD and Helm**, enabling automated reconciliation, repeatable deployments, and consistent environment promotion across Kubernetes workloads.
- Provisioned and managed infrastructure using **Terraform across multiple AWS regions**, importing previously manually managed resources into Infrastructure as Code to reduce configuration drift and improve lifecycle governance.
- Led an **AWS cost optimization initiative** with the DevOps team, reviewing resource utilization, decommissioning unused infrastructure, and standardizing instance recommendations ahead of Reserved Instance adoption, contributing to approximately **30% reduction in AWS costs**.
- Standardized platform observability using **Prometheus and Grafana** for metrics and alerting, **Jaeger** for distributed tracing, and **Sentry** for error monitoring across AWS EKS, Hetzner bare-metal Kubernetes, and legacy LXC environments.
- Drove observability instrumentation across business-critical **Rails and Flask services**, implementing OpenTelemetry tracing and Prometheus exporters to extend visibility into applications that previously lacked sufficient metrics and distributed tracing.
- Deployed a self-hosted **Jaeger platform on AWS EKS** using dedicated tainted node groups and an OpenTelemetry Collector pipeline, routing traces to Jaeger and Elasticsearch while isolating observability workloads from production application resources.
- Supported hybrid infrastructure spanning **AWS EKS, Hetzner bare-metal Kubernetes, and LXC**, using Ansible and Kubernetes tooling to troubleshoot deployment failures and maintain application and infrastructure reliability.
- Worked on connectivity between AWS and Hetzner environments, including **Site-to-Site VPN, routing, firewall configuration, DNS, and service reachability**, resolving cross-environment networking and application connectivity issues.
- Researched and documented Kubernetes multi-cluster connectivity between **Hetzner and AWS EKS using Submariner**, evaluating secure cross-cluster service discovery and communication patterns.
- Migrated authentication for self-hosted **Sentry from Okta to Microsoft Entra ID**, maintaining continuity of SSO access and identity management.
- Supported production incident response and root-cause analysis across Kubernetes, applications, databases, and networking, using metrics, tracing, and centralized logging to diagnose reliability and performance issues.
- Collaborated with Engineering, DevOps, and SRE teams to establish cloud-native architecture, observability, deployment, and operational standards across production environments.

Software Engineer / DevOps Engineer

Eneza Education **May 2019 – Dec 2024**

- Helped drive the modernization of production workloads from individually managed **AWS EC2 instances and Auto Scaling Groups to AWS EKS**, improving deployment consistency, scalability, resource utilization, and operational efficiency.
- Orchestrated modernization and integration of the core messaging platform with **Safaricom's Service Delivery Platform**, improving subscription, messaging, and billing workflows supporting mobile-based education services.
- Developed a reusable **Python integration package** for Safaricom and MTN Service Delivery Platforms via REST APIs, simplifying integration across subscription, SMS, and billing microservices.
- Engineered a high-throughput SMS gateway integrating with mobile network operator **REST APIs and SMPP**, improving the reliability and scalability of communication workflows.
- Built and operated asynchronous processing workflows using **RabbitMQ**, enabling decoupled and more resilient communication between distributed services.
- Implemented and maintained a highly available **Elasticsearch cluster** supporting business logging, search, monitoring, and operational analysis.
- Established and managed production **CI/CD pipelines using Jenkins and GitLab CI**, automating testing, deployment, and service delivery across multiple applications.
- Automated AWS infrastructure provisioning using **Terraform**, introducing reusable infrastructure patterns and environment separation to improve consistency, scalability, and maintainability.
- Deployed **Argo CD and Argo Rollouts** to introduce GitOps-based continuous and progressive delivery for Kubernetes workloads, improving deployment reliability and reducing manual release operations.
- Architected and operated **Kubernetes workloads on AWS EKS**, improving application scalability, resilience, deployment consistency, and

infrastructure utilization.
- Configured **NGINX Ingress** as a centralized entry point for microservices, simplifying routing and improving service exposure and traffic management.
- Improved observability using **Prometheus, ELK/Elasticsearch, and related monitoring tooling**, providing engineering teams with better visibility into service health and production performance.
- Performed data analysis using **Kibana dashboards and visualizations**, helping teams derive operational and business insights from application and messaging data.
- Maintained and enhanced legacy **PHP/Laravel applications** supporting the SMS learning platform, building and updating **business intelligence dashboards for B2C partners** to support operational reporting and business analysis.
- Contributed to infrastructure and application modernization initiatives that reduced technology-stack costs by approximately **65%** through cloud-resource optimization, architecture improvements, and removal of inefficient infrastructure.
- Led and participated in incident response and root-cause analysis, identifying recurring reliability issues and implementing improvements to platform stability and uptime.

Software Engineer

Hivisasa **Jul 2018 – Apr 2019**

- Led the design, development, and deployment of a **Python-based service replacing a legacy Java CMS**, improving maintainability and simplifying content-management workflows.
- Developed a Python microservice supporting the **writer reward system**, enabling automated reward workflows and improving contributor engagement.
- Collaborated on implementation of article-reaction functionality, extending user interaction capabilities across the publishing platform.
- Integrated writer-survey recruitment functionality into an **Angular application**, improving workflows used to onboard and engage content contributors.

Software Engineer

Pezesha **Feb 2018 – Jul 2018**

- Collaborated on the design and integration of microservices powering an **SMS-based digital lending platform**, supporting borrower onboarding and access to loan services through mobile channels.
- Worked with backend APIs and distributed services in a fintech environment where reliability, secure integrations, and transaction-oriented workflows were important.

Software Developer

Yum Deliveries **Mar 2017 – Feb 2018**

- Designed and implemented full-stack functionality using **Django, PostgreSQL, NGINX, and DigitalOcean**, supporting the development and operation of the food-delivery platform.
- Developed REST APIs supporting communication between backend services and mobile applications, improving application integration and feature delivery.
- Built business-intelligence dashboards visualizing restaurant, order, and user activity, enabling stakeholders to make more informed operational decisions.
- Implemented **JWT-based authentication for REST APIs**, strengthening application security and protecting backend resources.

Software Developer

Software Technologies Limited **Sep 2015 – Feb 2017**

- Developed backend and frontend functionality for enterprise applications using **PHP, Yii, SQL, and web technologies** across healthcare, recruitment, and HR management platforms.
- Implemented **role-based access control (RBAC)** using the Yii framework for healthcare and recruitment platforms, strengthening application authorization and access management.
- Collaborated with stakeholders to gather requirements and develop a greenfield **clinic-management application**, contributing across both frontend and backend implementation.
- Coordinated technical collaboration between **three remote developers in India and four local developers**, providing guidance on Yii framework practices and coding standards for an HRMS platform.
- Volunteered as a lecturer at the **Institute of Software Technologies**, sharing knowledge of PostgreSQL fundamentals with software-development students.

CONSULTING ENGAGEMENTS

Software Engineer / DevOps Engineer Consultant

AfricaSokoni **Jan 2020 – Jan 2023**

- Led the design and migration of Python microservices from manually managed infrastructure to **Infrastructure as Code**, improving deployment repeatability and supporting modernization of a legacy Joomla-based platform.
- Designed and deployed backend integrations using **Python microservices and AWS API Gateway**, improving scalability and separation between application components.

- Configured and maintained **Elasticsearch** for full-text search and infrastructure monitoring, improving search functionality and operational visibility.
- Integrated payment platforms including **iPay, Safaricom M-PESA, and Interswitch**, enabling reliable payment processing across the platform.
- Integrated **WhatsApp Business** capabilities to expand customer communication channels and improve digital engagement.

Software Engineer / DevOps Engineer Consultant

AfriSkul **Jun 2018 – Aug 2018**

- Developed and deployed a cross-platform educational technology application for video-based learning, integrating external APIs to improve content accessibility and user experience.

TECHNOLOGIES

Programming Languages: Python, Go, C#, JavaScript, TypeScript, SQL, PHP

Backend & APIs: Django, Flask, Rails, REST APIs, gRPC, Microservices, API Integrations

Cloud & Infrastructure: AWS, Amazon EKS, EC2, RDS, S3, Lambda, Route53, ElastiCache, Docker, Kubernetes, Terraform, Ansible, LXC, Hetzner

Platform Engineering & Delivery: Argo CD, Argo Rollouts, Helm, GitOps, Jenkins, GitLab CI/CD, GitHub Actions, NGINX Ingress, Istio, Apache APISIX

Observability & SRE: Prometheus, Grafana, OpenTelemetry, Jaeger, Kiali, Elasticsearch, Kibana, Sentry, Datadog, ELK, CloudWatch

Databases & Messaging: PostgreSQL, MySQL, Microsoft SQL Server, MongoDB, Redis, Elasticsearch, RabbitMQ, Kafka

Frontend: React.js, Angular, JavaScript/TypeScript

Networking & Security: IPSec, Site-to-Site VPN, mTLS, JWT, IAM, RBAC, API Authentication and Authorization, SSO, Microsoft Entra ID, Okta

Additional / Earlier Experience: Java, C++, jQuery, SOAP, Travis CI, PowerBuilder, Visual Basic